

Real-Time Motion Detection System Using I2C-Based MPU6050 Sensor

Introduction

This project implements a real-time motion detection system using an embedded microcontroller and an MPU6050 accelerometer sensor. The system reads acceleration data through the I2C communication protocol and detects motion based on predefined thresholds.

Objectives

- To interface MPU6050 sensor using I2C protocol
- To read accelerometer data (X, Y, Z axes)
- To implement motion detection logic
- To transmit data using UART communication

Components Used

- STM32F407VG Microcontroller
- MPU6050 Accelerometer & Gyroscope
- UART Communication Interface
- I2C Communication Protocol

System Design

I2C Communication

I2C is a two-wire communication protocol consisting of:

- SDA (data line)
- SCL (clock line)

It allows communication between the microcontroller and sensor.

Hardware Connections

MPU6050 STM32

VCC 3.3V

GND GND

SDA PB7

SCL PB6

Methodology

Sensor Initialization

The MPU6050 is initialized by disabling sleep mode using register 0x6B.

Data Acquisition

Acceleration values are read from registers starting at 0x3B.

Data Processing

High and low bytes are combined:

$Ax = (\text{HighByte} \ll 8) \mid \text{LowByte};$

Motion Detection

Motion is detected if:

$\text{if } (Ax > 15000 \mid \mid Ay > 15000 \mid \mid Az > 15000)$

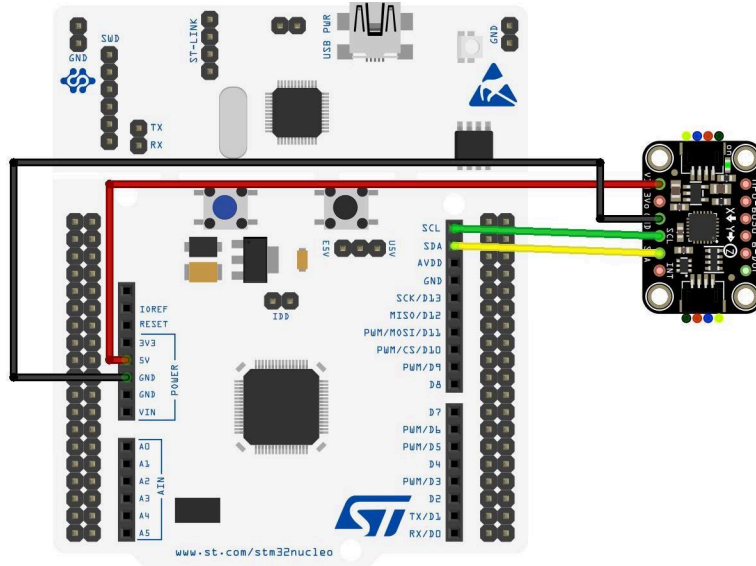
Step 5: UART Communication

Data is transmitted to a PC for monitoring.

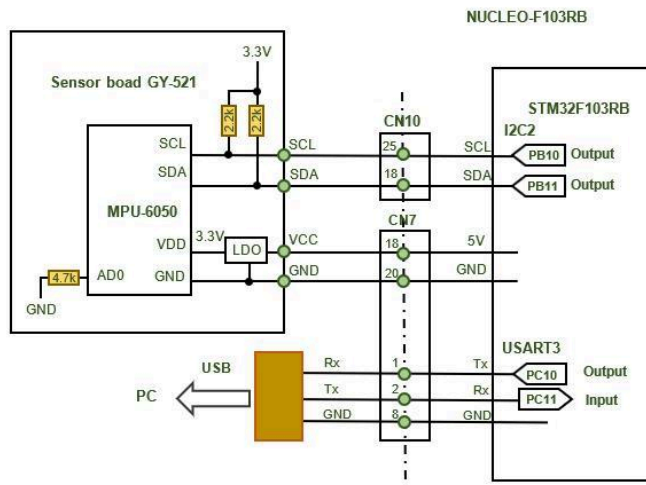
Simulation (Wokwi)

The Project was simulated in Wokwi using STM32 and maintaining the same IC2 communication logic.

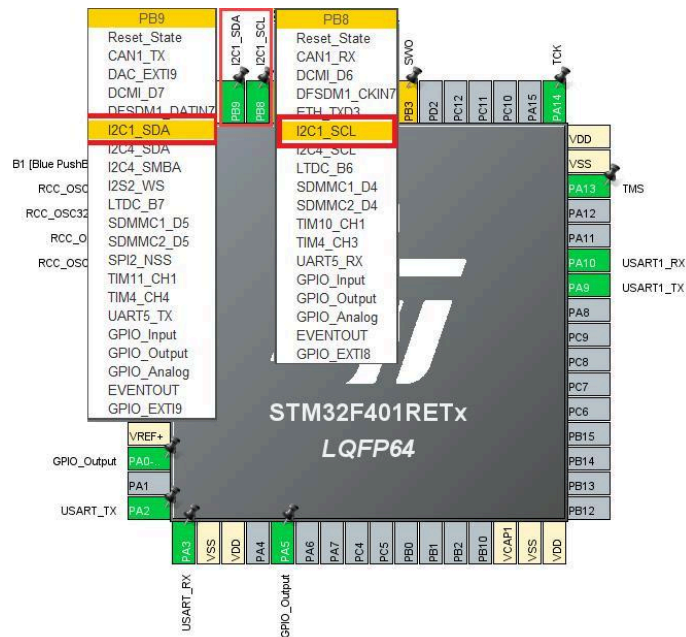
Bellow are the screenshots of the simulations and expected the output from the website:



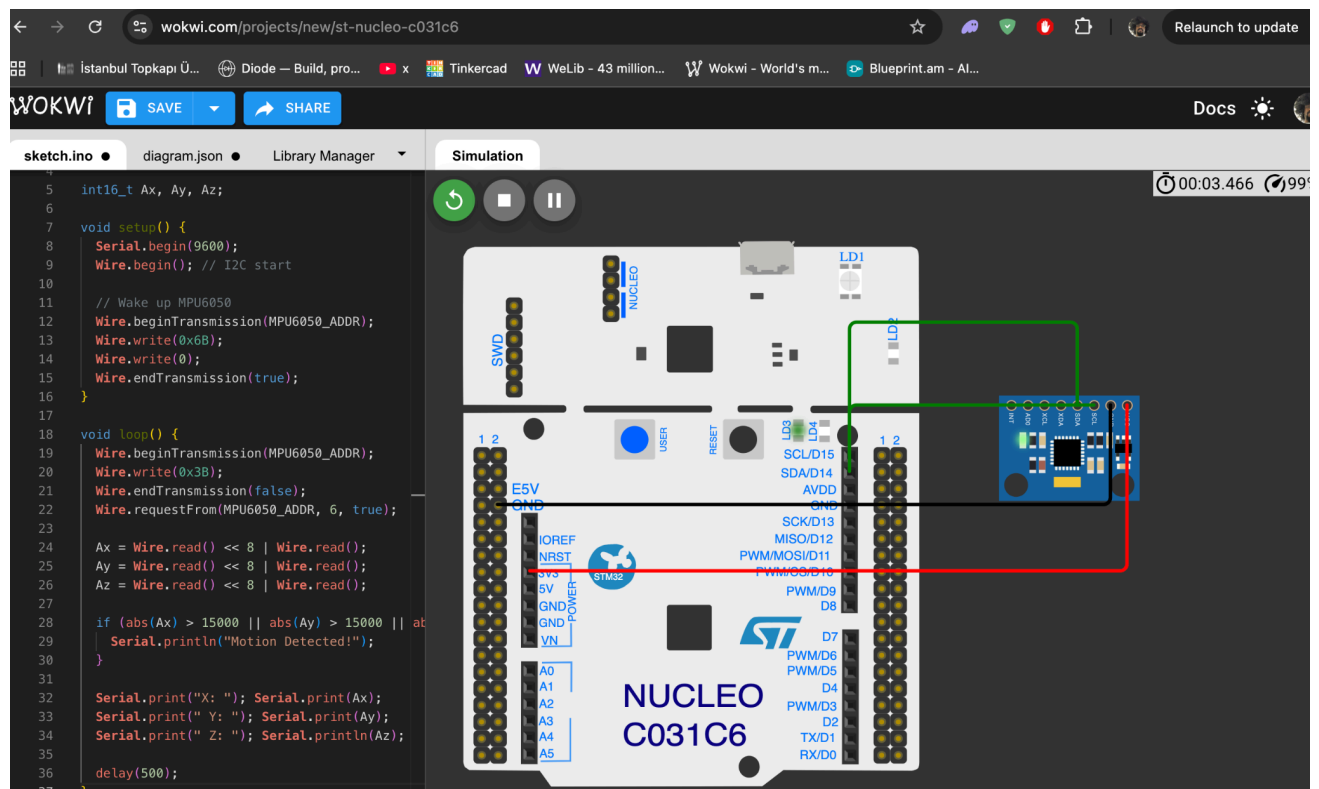
Circuit connection (Wokwi screen)



Schematic Diagram



Microcontroller STM32Cube IDE Configuration



Sim Screen Shot on Wokwi

Expected Output

Example:

X: 1200 Y: 300 Z: 16000

Motion Detected!

Results

- Accelerometer data was successfully read
- Real-time values were displayed on Serial Monitor
- Motion detection triggered correctly when threshold exceeded

Conclusion

The system successfully demonstrates real-time motion detection using I2C communication. The MPU6050 sensor provides accurate acceleration data, and motion detection logic works effectively.